American Computer Science League

2024-2025 ● Contest 3: Short Problems ● Classroom Division

1. Boolean Algebra

How many ordered pairs make the following Boolean expression TRUE?

$$A\,\overline{B} + \overline{A\,B} + A$$

A. 0
B. 1
C. 2
D. 3
E. 4

2. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{A\,B} \;+\; \overline{A(B + \overline{A})}$$

A. $\overline{A} + \overline{B}$
B. $A + \overline{A} * \overline{B}$
C. $\overline{A} + B$
D. $A + B$
E. $\overline{A} + A\overline{B}$

3. Data Structures

How many nodes have only 1 child in the binary search tree for:

EQUESTRIAN

A. 1
B. 2
C. 3
D. 4
E. 5

4. Data Structures

Given an initially empty queue and the following commands on the queue, what items are left from the rear to the front of the queue?

```
PUSH("F")
PUSH("E")
PUSH("N")
x = POP()
PUSH("C")
x = POP()
PUSH("I")
PUSH("N")
x = POP()
PUSH("G")
x = POP()
x = POP()
```

A. E F
B. N G
C. G F
D. G N
E. I N

5. Boolean Algebra

How many ordered triples make this expression TRUE?

$$A(\overline{B} + C) + \overline{AB} + C$$

A. 1
B. 3
C. 4
D. 5
E. 7

6. Boolean Algebra

Simplify this expression:

$$\overline{A}B(\overline{C + A}) + A\,B(\overline{\overline{A} + C})$$

A. $B + \overline{C}$
B. $\overline{A}B\overline{C}$
C. $A + \overline{B}C$
D. $B\overline{C}$
E. $A\overline{B}C$

7. Data Structures

List the nodes at depth 2 from left to right in the binary search tree for:

RUGBYSEVENS

A. B, E, N, S, Y
B. B, N, S, Y
C. E, N, S, Y
D. B, N, S, V
E. E, N, S

8. Data Structures

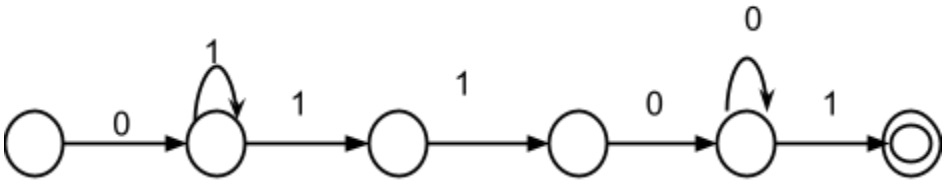
Given an initially empty stack and the following commands on the stack, what item will be popped next?

PUSH("A")
PUSH("R")
PUSH("C")
x = POP()
PUSH("H")
x = POP()
x = POP()
PUSH("E")
PUSH("R")
PUSH("Y")
x = POP()
x = POP()

A. A
B. C
C. E
D. R
E. H

9. FSAs

How many of the following strings satisfy the FSA below?



a. 01111101 b. 011101 c. 0111111

d. 0111001 e. 0110 f. 011110001

A. 1
B. 2
C. 3
D. 4
E. 5

10. Regular Expressions

Which of the following strings match the regular expression below?

[J-PT-V]*[^c^f-m]*[2468]

a. Jjp2 b. KULaazzz24 c. L6

d. TTOOB8 e. dstxyz4 f. 4

A. a, c, f
B. b, d, e
C. c, d, e, f
D. a, b
E. b, e, f